# Experiment 2

Develop a C program that uses the system calls open(), read(), write(), and close() to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system calls and identify the kernel services involve

```
#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>

#include <stdlib.h>


int main()

{

   int source, destination;

   char buffer[1024];

   int bytes;


   source = open("source.txt", O_RDONLY);


   if (source < 0)

   {

      perror("Error opening source file");

      return 1;
```

```
}

destination = open("destination.txt",
            O_WRONLY | O_CREAT | O_TRUNC,
            0644);

if (destination < 0)
{
  perror("Error creating destination file");
  close(source);
  return 1;
}

while ((bytes = read(source, buffer, sizeof(buffer))) > 0)
{
  write(destination, buffer, bytes);
}

close(source);
close(destination);

printf("File copied successfully.\n");

return 0;
```

}

Output:

```
ashwithreddy_earla@ASHWITHREDDYPC:~$ nano Experiment2.c
ashwithreddy_earla@ASHWITHREDDYPC:~$ echo "Hello, this is my Operating Systems experiment." > source.
ashwithreddy_earla@ASHWITHREDDYPC:~$ cat source.txt
Hello, this is my Operating Systems experiment.
```